

Internal Marks: 50
External Marks: 50

L	T	P	Cr
3	1	0	3.5

Course Objectives: The objective of this course is to give understanding on how current flows through the p-n junction and relating this phenomenon to the characteristics and operation of the diodes, bipolar and field-effect transistors. Students will learn the applications of diode as clipper, clamper and rectifier, operation of BJTs, FETs, behavior of BJTs at low frequency and working of SCR.

Section-A

Semiconductor Diodes: Qualitative analysis of a P-N junction as a diode, V-I characteristics and its temperature dependence, Break down diodes, Varactor diode, Tunnel diode, Semiconductor photo voltaic effect, Light emitting diodes.

Application of Diodes: Diodes as a clipper, Diode as a clamper, Sampling gate, Diode as a rectifier: Half wave rectifier, Full wave rectifier, Bridge rectifier, ripple factor, regulation, capacitor input filters, inductor input filter, L-type and π type-filters, Electronic Voltage regulator (Series and Shunt type).

Transistor Characteristics: Junction transistor current components, Transistor as an amplifier, Transistor construction: Common base, Common emitter, Common collector configuration, Various regions, photo transistor.

Transistor at Low frequencies: Graphical analysis of CE configuration two port devices and hybrid model, h-parameters, Conversion formulas for parameters of three configurations. Analysis of a transistor amplifier circuit using h-parameter, Miller theorem & its dual, comparison of amplifier configuration.

Section-B

Transistor Biasing and Thermal Stabilization: Need of biasing, Types of biasing circuits, Operating point, Bias stability, Stabilization against variation in I_{CO} , V_{BE} and β , Compensation techniques, Thermal run away, Thermal stability.

Field Effect Transistors: Junction field effect transistor, Pinch off voltage, JFET characteristics, and equivalent circuit, MOSFETS, their construction and characteristics, Biasing of JFETs.

Thyristors: Thyristor family, SCR-static and dynamic characteristics, Thyristor commutated Techniques, Two-transistor analogy, Triggering circuits, Thyristor applications.

ECEB2102T**ELECTRO-MAGNETIC FIELD THEORY****Internal Marks: 50**

L	T	P	Cr
3	1	0	3.5

External Marks: 50

Course Objectives: To enhance student's comprehensive capabilities in electromagnetic field theory by study the behavior of statics and time varying electric and magnetic field in a medium and transmission line.

Section-A

Concepts of vector algebra; Del operator, Curl, Divergence, Gradient, Laplacian Operator Gauss divergence theorem, Stokes theorem.

Electrostatic Fields: Review of Coulomb's law, Gaussian law, Laplace's and Poisson's equation in various coordinate systems, Effect of dielectric on capacitance, Boundary conditions at electric interfaces.

Electromagnetic Induction: Faraday's law, self and mutual inductance of coaxial cable, Energy stored in Electric and Magnetic fields.

Magnetic Fields: Interaction of currents and magnetic fields, Ampere's law, Magnetic vector potential, Boundary conditions at magnetic interfaces.

Section-B

Maxwell's Equation: Continuity equation, Concept of displacement current, Maxwell's equation in integral and differential form for static and time varying fields, Concept of Poynting's vector, Poynting's theorem, Power loss in a plane conductor.

Electromagnetic Waves: Wave equation, Propagation of uniform plane wave in free space and conducting medium, Polarization, Reflection and Refraction of plane waves, Brewster angle, Surface impedance.

Transmission Lines: Equivalent circuit of transmission lines, primary and secondary constants, transmission line equation, condition for loss-less and distortion-less line, loading of lines, quarter wave transformer, concept of VSWR and reflection coefficients, stub matching, Smith Chart and its use in transmission lines.

ECEB2103T ELECTRICAL AND ELECTRONIC INSTRUMENTATION

Internal Marks: 50

External Marks: 50

L	T	P	Cr
3	1	0	3.5

Course Objectives: The objective of the course is to provide a brief knowledge of measurements and measuring instruments related to engineering. The basic idea of this course is to give the sufficient information of measurements in any kind of industry viz. electrical, electronics, mechanical etc.

Section-A

Sensors and Transducers: Definition, classification, Basic principle & applications of Resistive, Inductive, Capacitive, Piezoelectric, Hall-Effect, photo electric transducer, load cell and Digital Transducers, selection of sensors, Recent trends and smart sensors, Instrumentation Amplifiers.

Voltage & Current Measurement: Measurement of DC & AC current & voltage; Basics of indicating instruments. Principle of operation and working of PMMC, dynamometer, Moving Iron, Rectifier and Thermocouple Instruments. VTVM, TVM-Ramp Type, Integrating Type, Potentiometric Type, Successive approximation Type, continuous Balance Type.

Section-B

RLC Measurements: Measurement of DC medium resistance (Wheatstone method.) AC Wheatstone Bridge, Maxwell, Hay, Wien and Schering Bridge, Q Meter.

CRO & Display Devices: Basic Block diagram, functions of Blocks & Basic CRO circuits, Measurement of voltage, current, frequency and phase angle, Dual beam oscilloscope, digital display devices: construction & working principle-LED, LCD, Magnetic Recorders.

Measurement of Power: Measurement of Power using Bolometer Bridge method, Measurement of Power & Power factor using one wattmeter, two wattmeter method.

Text Books: -

1. Electrical & electronic Measurement & Instrumentation by A.K. Sawhney, Dhanpat Rai & Sons.
2. Electronic Instrumentation & Measurement Techniques by W.D. Cooper, PHI, New Delhi.
3. Electronic Measurement by Terman & Pettit: Tata McGraw Hall.
4. Electronic Instrumentation by H.S. Kalsi.
5. Handbook of Transducers BY H.N. Norton, Prentice Hall Inc.

ECEB2104T**MANAGEMENT PRACTICES & ORGANIZATIONAL
BEHAVIOUR****Internal Marks: 50****L T P Cr****External Marks: 50****3 0 0 3.0**

Course Objectives: The course prepares the students for successful professional life by inculcating in them the abilities to understand the human interactions in an organization, find what can drive human behaviour and influence them to get better results for attaining business planning, organizing, staffing, directing, coordinating and controlling the activities of business enterprise.

Section – A

Introduction to Management: Definition, Importance and functions of Management. Theories of Management; Classical, Neo-classical and Modern. Planning: Nature of planning, planning process, types of plans, Importance and Limitations of Planning. Introduction to MBO (Management by Objectives), Social responsibility of business.

Decision Making: Importance and Process. Organization: Process of Organizing, Organizing Principles, Organization Chart, Authority and Responsibility relationship, Steps in Delegation of Authority.

Communication: Process, channels, medium of communication, communication barriers. Controlling: Steps, types of control system, essentials of effective control system.

Section – B

Organizational Behavior: Concept, features and importance. Personality: determinants and development. Role of Values and Attitudes in individual's behavior. The concept of motivation and its theories. Perception: Concept, Process, Errors in perceptual accuracy, Role of perception in decision making.

Learning: Classical and Operant conditioning theory, Reinforcement-kinds and administration. Concept of group dynamics. Leadership theories and styles.

Organizational Conflict: Concept, Dimensions, conflict management techniques. Introduction to concept of power and politics in work related organization. Organization culture and effectiveness.

Internal Marks: 50**External Marks: 50**

L	T	P	Cr
3	1	0	3.5

Course Objectives: To become familiar with object-oriented programming concepts using C++ and be able to apply these concepts in solving diverse range of applications.

Section – A

Introduction: Review of basic concepts (Data types, tokens, operators, arrays, strings, structure). Concepts of OOPS, Comparison between procedural and OOPS, Elements of OOPS, its paradigm, its merits and demerits, functions, function overloading and function.

Classes and Objects: Classes, creating a class, accessing class members, memory allocation for objects, array of objects, static objects, friend class, empty classes, nested classes, abstract classes and container class.

Constructor and Destructor: Its need, Types of constructor- Default, Copy, Multiple, Dynamic, Destructor, Constructor and Destructor with static members, Order of execution of constructor and destructor.

Operator Overloading and type Conversion: Definition, Rules for overloading, overloading of unary and binary operators, Overloading NEW and DELETE operators, Type conversion- Basic type to class type, class type to basic type, class to class type.

Section – B

Inheritance: - Virtual base class, types of inheritance, typing conversions and visibility.

Polymorphism: - Virtual functions, Virtual destructors, Concept of binding-early and late, Virtual functions, Pure virtual functions, Polymorphism.

Exception Handling: Basics of exception handling, Exception handling mechanisms, throwing mechanisms, Catching mechanisms

Pointers and Dynamic Memory Management: Understanding pointers, accessing address of a variable, declaring and initialization, pointer arithmetic, pointer-to-pointer, pointer-to-function, pointer-to-array, array of pointers, pointer-to-object, these pointers, null pointers, dangling pointers and memory leak.

Managing Data Files: File streams, Hierarchy of file streams, Reading/Writing of files, accessing records randomly and sequentially and updating files.

ERSB2101Q**Environmental and Road Safety Awareness**

Time Allowed: 3 hours
Internal: 30 Marks

Total Marks: 100
External: 70 Marks

Instructions

- a) This will cover only preliminary and basics of the subject and the paper will be set accordingly.
- b) The External question Paper will consist of three sections A, B and C. Sections A and B will have four questions each from the respective sections of the syllabus. Each questions shall carry 10 marks. Section C will consist of 10 short answer type questions of 3 marks each. Candidates are required to attempt any two questions each from section A and section B. Section C is compulsory.

Section – I

Unit 1 : The multidisciplinary nature of environmental studies. Definition, scope and importance

- Concept of Biosphere – Lithosphere, Hydrosphere, Atmosphere.
- Need for public awareness (6lectures)

Unit – 2 Natural Resources – Renewable and non-renewable resources.

- Natural resources and associated problems.
 - a) Forest resources: use and over exploitation, deforestation and its impact.
 - b) Water resources; use and overutilization of surface and ground water and its impact.
 - c) Mineral resources: use and effects on environment on over exploitation.
 - d) Food resources: Effects modern agriculture, fertilizer-pesticide problem, water logging and salinity.
 - e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy resources.
 - f) Role of an individual in conservation of natural resources for sustainable development.

(7 lectures)

Unit 3: Ecosystems

- Ecosystem and its components: Definition, structure and function; producer, consumer and decomposer.
- Types of Ecosystem (Introduction only)
- Food Chains, food web and ecological pyramids (6 lectures)

Unit – 4: Biodiversity and conservation

- Introduction – Definition: genetic, species and ecosystem diversity, value of biodiversity.
- Hot spots of biodiversity
- Threats to biodiversity: habitat loss, pooceting of wildlife, man-wildlife conflicts.
- Endangered and endemic species of India.
- Conservation of Biodiversity. (6 lectures)

Section – II

Units 5 : Environmental Pollution

- Definition, causes, effects and control measures of
 - a) Air pollution
 - b) Water pollution
 - c) Soil pollution
 - d) Marine pollution
 - e) Noise pollution
 - f) Thermal pollution
 - g) Nuclear hazard
- Role of an individual in prevention of pollution.
- Solid waste management: vermicomposting.
- Disaster management: Floods, earthquake, cyclone and landslides (7 lectures)

Unit 6: Social Issues and the Environment

- Urban problems related to energy.
- Water conservation rain water harvesting, water shed management.
- Resettlement and rehabilitation of people: its problems and concerns.
- Climate changes, global warming, acid rain, ozone layer depletion.
- Consumerism and waste products.
- Population explosion – Family welfare programme lectures) (6

Unit 7: Introduction to Environmental Protection Laws in India

- Environmental Protection Act.
- Air (Prevention and control of pollution) Act.
- Water (Prevention and Control of pollution) Act.
- Wild life Protection Act.
- Forest Conservation Act.
- Issues involved in the enforcement of environmental legislation. (6 lectures)

Unit 8: Road safety Awareness

- Concept and significance of Road safety.
- Traffic signs.
- Traffic rules.
- Traffic Offences and penalties.
- How to obtain license.
- Role of first aid in Road Safety. (6 lectures)

Suggested Readings:

1. Carson, R. 2002. Silent Spring, Houghton Mifflin Harcourt.
2. Gadgil, M., & Guha, R. 1993. This Fissured Land: An Ecological History of India. Univ. of California Press.
3. Gleeson, B. and Low, N. (eds.) 1999. Global Ethics and Environment, London, Routledge.
4. Gleick, P. H. 1993. Water in Crisis. Pacific Institute for Studies in Dev. Environment & Security. Stockholam Env. Institute, Oxford Univ. Press.
5. Groom, Martha J., Gary K. Meffe, and Carl Ronald Carroll. Principles of Conservation Biology. Sunderland: Sinauer Associates, 2006.
6. Grumbine, R. Edward, and Pandit, M.K. 2013. Threats from India's Himalays dams. Science, 339:36-37.
7. McCully, P. 1996. Rivers no more: the environmental effects of dams (pp.29-64). Zed Books.
8. McNeill, John R. 2000. Something New Under the Sun: An Environmental History of the Twentieth Century.
9. Odum, E.P., H.T. & Andrews, J. 1971. Fundamentals of Ecology. Philadelphia: Saunders.
10. Pepper, I.L., Gerba, C.P. & Brusseau, M.L. 2011. Environmental and Pollution Sciences. Academic Press.
11. Rao, M.N. & Datta, A.K. 1987. Waste Water Treatment. Oxford and IBH Publishing Co. Pvt. Ltd.
12. Raven, P.H., Hassenzahl, D.M. & Berg, L.R. 2012, Environment. 8Th edition. John Wiles & Sons.
13. Rosencranz, A., Divan, S., & Nobie, M.L. 2001. Environmental law and policy in India. Tripathi 1992
14. Sengupta, R. 2003. Ecology and economics: An approach to sustainable development. OUP.
15. Singh, J.S., Singh, S.P. and Gupta, S.R. 2014. Ecology, Environmental Science and Conservation. S. Chand Publishing, New Delhi.
16. Sodhi, N.S. Gibson, L. & Raven, P.H. (eds). 2013. Conservation Biology: Voices from the Tropics. John Wiley & Sons.
17. Thapar, V. 1998. Land of the Tiger: A Natural History of the Indian Subcontinent.
18. Warren, C.E. 1971. Biology and Water Pollution Control. WB Saunders.

DMPB2102Q**DISASTER MANAGEMENT AND PREPAREDNESS**Time Allowed: 3 hours
Internal: 50 MarksTotal Marks: 100
External: 50 Marks**Section – A**

Understanding Disasters: Understanding the Concepts and definitions of Disaster, Hazard, Vulnerability, Risk, Capacity – Disaster and Development, and disaster management

Types, Trends, Causes, Consequences and Control of Disasters: Geological Disasters (earthquakes, landslides, tsunami, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunder-storms, hail storms, avalanches, droughts, cold and heat waves) Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear) and Manmade Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters) Global Disaster Trends – Emerging Risks of Disasters – Climate Change and Urban Disasters

Disaster Management Cycle and Framework: Disaster Management Cycle – Paradigm Shift in Disaster Management Pre-Disaster – Risk Assessment and Analysis, Risk Mapping, zonation and Micro-zonation, Prevention and Mitigation of Disasters, Early Warning System; Preparedness, Capacity Development; Awareness During Disaster – Evacuation – Disaster Communication – Search and Rescue – Emergency Operation Centre – Incident Command System – Relief and Rehabilitation – Post-disaster – Damage and Needs Assessment, Restoration of Critical Infrastructure – Early Recovery – Reconstruction and Redevelopment; IDNDR, Yokohama Strategy, Hyogo Framework of Action

Section – B

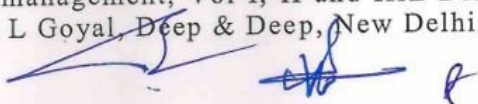
Disaster Management in India: Disaster Profile of India – Mega Disasters of India and Lessons Learnt Disaster Management Act 2005 – Institutional and Financial Mechanism National Policy on Disaster Management, National Guidelines and Plans on Disaster Management; Role of Government (local, state and national), Non-Government and Inter-Governmental Agencies

Applications of Science and Technology for Disaster Management: Geo-informatics in Disaster Management (RS, GIS, GPS and RS) Disaster Communication System (Early Warning and Its Dissemination) Land Use Planning and Development Regulations Disaster Safe Designs and Constructions Structural and Non-Structural Mitigation of Disasters S&T Institutions for Disaster Management in India

Suggested Areas for Project and Assignments: Study of Recent Disasters (at local, state and national level) And Preparation of Disaster Risk Management Plan of an Area or Sector Role of Engineers in Disaster Management

Suggested Readings:

1. Coppola D P, 2007. Introduction to International Disaster Management, Elsevier Science (B/H), London.
2. Manual on natural disaster management in India, M C Gupta, NIDM, New Delhi
3. An overview on natural & man-made disasters and their reduction, R K Bhandani, CSIR, New Delhi
4. World Disasters Report, 2009. International Federation of Red Cross and Red Crescent, Switzerland
5. Encyclopedia of disaster management, Vol I, II and III Disaster management policy and administration, S L Goyal, Deep & Deep, New Delhi, 2006

ECEB2107P**ELECTRONIC DEVICES LAB****Internal Marks: 50****External Marks: 50**

L	T	P	Cr
0	0	2	1.0

Course Objectives:

1. To identify and test various electronic components.
2. To use CRO for various measurements.
3. To plot the V-I characteristics of diodes and transistors.

LIST OF EXPERIMENTS

1. Study of h-Parameters of CE transistor.
2. Study of h-Parameters of CB transistor.
3. Study of V-I characteristics of Photodiode.
4. Study of V-I characteristics of photo transistor.
5. Study of V-I characteristics of JFET.
6. Study of VI characteristics of MOSFET.
7. Study of Clipping & clamping CKT using Diode.
8. Study of V-I characteristics of SCR.
9. Study of Diode as rectifier.
10. Study of different filters.
11. Study of V-I characteristics of DIAC
12. Study of V-I characteristics of TRIAC.
13. Study the applications of Thyristor.

Special Note: In addition to above experiments, every student has to prepare a working project/kit/hardware or software model based on the above syllabus/experiments. 20% weightage (internal (10%) + external (10%)) will be given for this during evaluations.

Course Outcomes: At the end of this course, the student will be able to:

1. Measure voltage, frequency and phase of any waveform using CRO.
2. Verify the rectifier circuits using diodes and implement them using hardware.
3. Analyze the characteristics of different electronic devices such as diodes, transistors etc., and simple circuits like rectifiers, filters etc.
4. Analyze the concepts of SCR and observe its characteristics.

ECEB2108P OBJECT ORIENTED PROGRAMMING USING C++ LAB**Internal Marks: 50****External Marks: 50**

L	T	P	Cr
0	0	2	1.0

Course Objectives:

1. To familiarize students with object-oriented concepts and their implementation in C++.
2. To facilitate students with the skills required to solve problems using object-oriented concepts.

LIST OF EXPERIMENTS

1. Write a program to illustrate the working of different iterative controls.
2. Write a program to illustrate the use of conditional statements.
3. Write a program to illustrate difference between call by reference and call by value.
4. Write a program to illustrate use of classes and objects.
5. Write a program to illustrate the use of constructors and destructors.
6. Write a program to illustrate the use operator overloading.
7. Write a program to illustrate the use of inheritance and type of inheritance.
8. Write a program to illustrate the use virtual function.
9. Write a program to illustrate the use runtime polymorphism.
10. Write a program to illustrate the use of File Stream operation.
11. Create a class **Time** with members hours, minutes, seconds. Add two class objects and display time.
12. Write a program to perform addition of two complex numbers using constructor overloading.
13. Write a program to find the greatest of two given numbers in two different classes using friend function.
14. Implement a class **String** with the following:
 - Overload + operator to carry out the concatenation of strings.
 - Overload = operator to carry out string copy.
 - Overload <= operator to carry out the comparison of strings.
 - Function to display the length of a string.
 - Function *tolower()* to convert upper case letters to lower case.
 - Function *toupper()* to convert lower case letters to upper case.
15. Create a class called **List** with two pure virtual function *store()* to save a value and *retrieve()* to get a saved value from the list. Derive two classes **Stack** and **Queue** from it and override store and retrieve.
16. Write a program to define the function template for calculating the square of given numbers with different data types.
17. Write a program to perform the deletion of white spaces such as horizontal tab, vertical tab, space, line feed, new line and carriage return from a text file and store the result in another file.
18. Write a program to raise an exception if any attempt is made to refer to an element whose index is beyond the array size.

ECEB2110P ELECTRONICS AND INSTRUMENTATION LAB

Internal Marks: 50

L

T

P

C

External Marks: 50

0

0

2

1.0

Course Objectives:

1. To introduce the fundamentals of Electronics Instruments and Measurement providing an in-depth understanding of Measurement errors, Bridge measurements, CRO, Function Generator and Analyzer, Display devices, Data acquisition systems and transducers.
2. To address the underlying concepts and methods behind Electronics measurements.

LIST OF EXPERIMENTS

1. To study the function of Function Generator and all its waveforms with the help of CRO.
2. To Study the operation and technical specifications of Multimeter.
3. Measurement of wind speed using Anemometer
4. To measure strain with the help of Strain Gauge.
5. Study & Measurement of mechanical Vibrations
6. To Study the operation and technical Specification of Thermocouple.
7. Study of Voltmeters and Ammeters
8. Study & Measurement of speed (RPM) using L.D.R
9. Study the working of Dead Weight Tester for Pressure Measurement
10. Study & Measurement of Force using Load Cell

Course Outcomes: At the end of this course, the student will be able to:

1. Identify different measuring instruments for the measurement of various electronic parameters.
2. Select various transducers for the measurement of physical quantities like temperature, pressure, distance and displacement.
3. Compute the errors present in measuring instruments and calibrate them.
4. Examine AC bridges for the measurement of inductance, capacitance and frequency.